Wilhan *Nunes*

PH.D. IN CHEMISTRY

Research Staff Associate II · Computational Metabolomics

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La Jolla, CA 92093

PROFILE

Research Staff Associate II specializing in computational metabolomics at UC San Diego. My work spans the creation and curation of microbial metabolite databases, development of mass spectrometry-based analytical workflows, and web-based tools that bridge experts and non-experts in the metabolomics field. With over a decade of academic experience across multiple institutions in Brazil and the US, I bring a track record in both research and teaching.

RESEARCH PROJECTS

2024 — Present

Computational Metabolomics & Resource Development

Dr. Pieter C. Dorrestein · Skaggs School of Pharmacy, UC San Diego

- **Microbial metabolite knowledgebase:** Development of the Collaborative Microbial Metabolite Center knowledgebase (CMMC-kb), integrating spectral and structural data of microbial metabolites with biological information including producers, bioactivity, and biosynthesis.
- **Open science tools:** Designed and coded the user interface for MetaboApps — a suite of modular web applications delivering accessible, standardized GNPS-based metabolomics workflows to the broader scientific community.

2015 — 2018

Synthesis and Thermoanalysis of Lanthanide/Yttrium Isonicotinates

Doctoral thesis · Dr. Flavio J. Caires & Dr. Massao Ionashiro · UNESP, Brazil

Synthesized and characterized Ln(III)/Y(III) isonicotinate complexes and evaluated their thermal behavior using TGA, DSC, XRD, and wet-chemistry analysis. Funded by CAPES doctoral scholarship.

2013 — 2015

Synthesis and Thermoanalysis of Lanthanide Isophthalates

Master's dissertation · Dr. Elias Y. Ionashiro · UFG, Brazil

Prepared and characterized isophthalate complexes of La(III), Ce(III), Pr(III), Nd(III), Sm(III), Eu(III), and Gd(III), correlating structure with thermal stability via TGA, DSC, and XRD. Supported by CAPES scholarship.

2009 — 2013

Low-cost Thin-Film Fabrication & Instrumentation

Undergraduate research · Dr. George Barbosa da Silva · UFMT, Brazil

Fabricated SnO_2 conductive glass electrodes via spray-pyrolysis and designed dip-coating and spin-coating apparatus using low-cost electronics. Funded annually by CNPq research scholarship.

PAST PROFESSIONAL EXPERIENCE

2022 — 2023

Chemistry Professor

Federal Institute of Education, Science and Technology of São Paulo (IFSP) · Ilha Solteira, SP, Brazil

Delivered Chemistry for Technical Degree tracks in Civil Construction Design and Building.

2018 — 2021

Chemistry Professor

Federal Institute of Education, Science and Technology of Rondônia (IFRO) · Ji-Paraná, RO, Brazil

Led courses in General, Physical, Analytical & Organic Chemistry (Licentiate) and General & Physical Chemistry (Technical tracks).

Feb - Jul 2018

Temporary Chemistry Professor

Federal University of Mato Grosso (UFMT) · Barra do Garças, MT, Brazil

Lectured Physical & Analytical Chemistry across Chemistry, Pharmacy, and Food Engineering programs.

Aug 2017 - Jan 2018

Chemistry Professor Internship

Faculty of Sciences, UNESP (Chemistry Department) · Bauru, SP, Brazil

Led experimental classes in the General and Inorganic Chemistry Laboratory for the Physics undergraduate program.

SKILLS

COMPUTATIONAL	Big data mining · Web application development for metabolomics · Workflow automation · Nextflow · Docker · Git (branching, rebase, submodules) · Database development · MS/MS data acquisition & analysis · Python · R · Bash · Arduino
WET LAB	Sample preparation for LC-MS metabolomics · Chromatographic method development, optimization, and validation
SOFTWARE	GNPS · MZmine · ChemDraw · Cytoscape · Inkscape

LANGUAGES	Portuguese (native) · English (fluent) · French (basic) · Spanish (basic)
SOFT SKILLS	Teamwork · Leadership · Scientific communication · Interpersonal skills

PUBLICATIONS

2026

Mohanty I., Xing S., Castillo V., Agongo J., Patan A., El Abiead Y. et al.
MS/MS Mass Spectrometry Filtering Tree for Bile Acid Regio- and Stereoisomer Annotation
Analytical Chemistry, v.98, p. 4571–4584. DOI ↗

Xing S., Patan A., Agongo J., Gouda H., Charron-Lamoureux V., El Abiead Y. et al.
Navigating the conjugated metabolome PREPRINT
. DOI ↗

Mannochio-Russo H., Ferreira P.C., Kvitne K.E., Patan A., Deleray V., Agongo J. et al.
Pan-Metabolomics Repository Mapping of the Carnitine Landscape PREPRINT
bioRxiv. DOI ↗

Mannochio-Russo H., Gonçalves Nunes W.D., Xing S., de Oliveira F., Caraballo-Rodríguez A.M., Portal Gomes P.W. et al.
Community Curation of Microbial Metabolites Enables Biological Insights of Metabolomics Data PREPRINT
bioRxiv. DOI ↗

2025

Patan A., Xing S., Charron-Lamoureux V., Hu Z., Deleray V., Agongo J. et al.
Charting the Undiscovered Metabolome with Synthetic Multiplexing PREPRINT
. DOI ↗

Dorrestein P., Gouda H., Agongo J., Kelly P., Sala-Climent M., Nunes W.G. et al.
Learning molecular fingerprints of foods to decode dietary intake PREPRINT
. DOI ↗

Mannochio-Russo H., Gonçalves Nunes W.D., Zhao H.N., Kvitne K.E., Xing S., Gouda H. et al.
Bridging Complexity and Accessibility in Metabolomics with MetaboApps PREPRINT
. DOI ↗

Mannochio-Russo H., Charron-Lamoureux V., van Faassen M., Lamichhane S., Gonçalves Nunes W.D., Deleray V. et al.
The microbiome diversifies long- to short-chain fatty acid-derived N-acyl lipids
Cell, v.188, p. 4154–4169.e19. DOI ↗

Flores Ramos S., Siguenza N., Zhong W., Mohanty I., Lingaraju A., Richter R.A. et al.
Metatranscriptomics uncovers diurnal functional shifts in bacterial transgenes with profound metabolic effects
Cell Host & Microbe, v.33, p. 1057–1072.e5. DOI ↗

Mannochio-Russo H., Delolo F.G., Silva R.C.F., da Silva F.L.F., Moreira E.A., Nascimento V. et al.
PERFIL DOS JOVENS DOUTORES EM QUÍMICA NO BRASIL: DESAFIOS, DISPARIDADES E PERSPECTIVAS
Química Nova. DOI ↗

2023

Mannochio-Russo H., Nunes W.D.G., Almeida R.F., Albernaz L.C., Espindola L.S., Bolzani V.S.
Old Meets New: Mass Spectrometry-Based Untargeted Metabolomics Reveals Unusual Larvicidal Nitropropanoyl Glycosides from the Leaves of Heteropterys umbellata
Journal of Natural Products, v.86, p. 621–632. DOI ↗

2022

Mannochio-Russo H., de Almeida R.F., Nunes W.D.G., Bueno P.C.P., Caraballo-Rodríguez A.M., Bauermeister A. et al.
Untargeted Metabolomics Sheds Light on the Diversity of Major Classes of Secondary Metabolites in the Malpighiaceae Botanical Family
Frontiers in Plant Science, v.13. DOI ↗

Silva D.H.S., Mannochio-Russo H., Lago J.H.G., Bueno P.C.P., Medina R.P., Bolzani V.d.S. et al.
Bioprospecting as a strategy for conservation and sustainable use of the Brazilian Flora
Biota Neotropica, v.22. DOI ↗

2020

Gonçalves Nunes W.D., Mannochio Russo H., da Silva Bolzani V., Caires F.J.

Thermal characterization and compounds identification of commercial Stevia rebaudiana Bertoni sweeteners and thermal degradation products at high temperatures by TG–DSC, IR and LC–MS/MS

Journal of Thermal Analysis and Calorimetry, v.146, p. 1149–1155. DOI ↗

2019

Zangaro G.A., Carvalho A.C., Ekawa B., do Nascimento A.L., Nunes W.D., Fernandes R.P. et al.

Study of the thermal behavior in oxidative and pyrolysis conditions of some transition metals complexes with Lornoxicam as ligand using the techniques: TG-DSC, DSC, HSM and EGA (TG-FTIR and HSM-MS)

Thermochimica Acta, v.681, p. 178399. DOI ↗

Nunes W.D.G., Teixeira J.A., do Nascimento A.L.C.S., Ekawa B., Ionashiro M., Caires F.J.

Thermal analysis (TG–DSC, DSC–microscopy and EGA) and characterization of heavy trivalent lanthanides and yttrium isonicotinates

Journal of Thermal Analysis and Calorimetry, v.138, p. 309–319. DOI ↗

2018

Ekawa B., Nunes W.D.G., Teixeira J.A., Cebim M.A., Ionashiro E.Y., Junior Caires F.

New complexes of light lanthanides with the valsartan in the solid state: Thermal and spectroscopic studies

Journal of Analytical and Applied Pyrolysis, v.135, p. 299–309. DOI ↗

Nascimento A., Parkes G., Ashton G., Fernandes R., Teixeira J., Nunes W. et al.

Thermal analysis in oxidative and pyrolysis conditions of alkaline earth metals picolinates using the techniques: TG-DSC, DSC, MWTA, HSM and EGA (TG-DSC-FTIR and HSM-MS)

Journal of Analytical and Applied Pyrolysis, v.135, p. 67–75. DOI ↗

Nunes W.D.G., Teixeira J.A., Ekawa B., do Nascimento A.L.C.S., Ionashiro M., Caires F.J.

Mn(II), Fe(II), Co(II), Ni(II), Cu(II) and Zn(II) transition metals isonicotinate complexes: Thermal behavior in N2 and air atmospheres and spectroscopic characterization

Thermochimica Acta, v.666, p. 156–165. DOI ↗

Costa B.A., Nunes W.D.G., Bembo L.H., Siqueira A.B., Caires F., Leles M.I.G. et al.

Study of thermoanalytical behavior of heavier lanthanides terephthalates in air atmosphere

Journal of Thermal Analysis and Calorimetry, v.134, p. 1205–1210. DOI ↗

Nunes W.D.G., do Nascimento A.L.C.S., Moura A., Gaglieri C., Vallim G.B., Nascimento L.C. et al.

Thermal, spectroscopic and antimicrobial activity characterization of some norfloxacin complexes

Journal of Thermal Analysis and Calorimetry, v.132, p. 1077–1088. DOI ↗

Russo H.M., Nunes W.D.G., da Silva Bolzani V., Ionashiro M., Caires F.J.

Thermoanalytical and spectroscopic characteristics of young and old leaves powder and methanolic extracts of Niedenzuella multiglandulosa

Journal of Thermal Analysis and Calorimetry, v.132, p. 771–776. DOI ↗

2017

Silva R.d.C.d., Teixeira J.A., Nunes W.D.G., Zangaro G.A.C., Pivatto M., Caires F.J. et al.

Resveratrol: A thermoanalytical study

Food Chemistry, v.237, p. 561–565. DOI ↗

Teixeira J., Nunes W., Fernandes R., do Nascimento A., Caires F., Ionashiro M.

Thermal behavior in oxidative and pyrolysis conditions and characterization of some metal p-aminobenzoate compounds using TG–DTA, EGA and DSC-photovisual system

Journal of Analytical and Applied Pyrolysis, v.128, p. 261–267. DOI ↗

VIII Simpósio de Análise Térmica - Ponta Grossa/PR, v.8. DOI ↗

Caires F.J., Nunes W.D.G., Gaglieri C., Nascimento A.L.C.S.d., Teixeira J.A., Zangaro G.A.C. et al.

Thermoanalytical, Spectroscopic and DFT Studies of Heavy Trivalent Lanthanides and Yttrium(III) with Oxamate as Ligand

Materials Research, v.20, p. 937–944. DOI ↗

R. Silva F.P., Viana B.F., Santos J.P., Nunes W.D., Teixeira J.A., Ionashiro E.Y.

SÍNTESE, CARACTERIZAÇÃO E ESTUDO TERMOANALÍTICO DOS ISOFTALATOS DE Mg (II), Ca (II), Sr (II), Ba (II)

Brazilian Journal of Thermal Analysis, v.6, p. 19–27. DOI ↗

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do Nascimento A.L.C.S., Teixeira J.A., Nunes W.D.G., Gomes D.J.C., Gaglieri C., Treu-Filho O. et al.

Thermal behavior of glycolic acid, sodium glycolate and its compounds with some bivalent transition metal ions in the solid state

Journal of Thermal Analysis and Calorimetry, v.130, p. 1463–1472. DOI ↗

2016

Teixeira J., Nunes W., do Nascimento A., Colman T., Caires F., Gálico D. et al.

Synthesis, thermoanalytical, spectroscopic study and pyrolysis of solid rare earth complexes (Eu, Gd, Tb and Dy) with p-aminobenzoic acid

Journal of Analytical and Applied Pyrolysis, v.121, p. 267–274. DOI ↗

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do Nascimento A., Teixeira J., Nunes W., Campos F., Treu-Filho O., Caires F. et al.

Thermal behavior, spectroscopic study and evolved gas analysis (EGA) during pyrolysis of picolinic acid, sodium picolinate and its light trivalent lanthanide complexes in solid state

Journal of Analytical and Applied Pyrolysis, v.119, p. 242–250. DOI ↗

Nunes W.D.G., Teixeira J.A., do Nascimento A.L.C.S., Caires F.J., Ionashiro E.Y., Ionashiro M.

A comparative study on thermal behavior of solid-state light trivalent lanthanide isonicotinates in dynamic dry air and nitrogen atmospheres

Journal of Thermal Analysis and Calorimetry, v.125, p. 397–405. DOI ↗

Teixeira J., Nunes W., Colman T., do Nascimento A., Caires F., Campos F. et al.

Thermal and spectroscopic study to investigate p-aminobenzoic acid, sodium p-aminobenzoate and its compounds with some lighter trivalent lanthanides

Thermochimica Acta, v.624, p. 59–68. DOI ↗

EDUCATION

Doctor of Philosophy in Chemistry
2018 - 2022

Institute of Chemistry, São Paulo State University — UNESP · Araraquara, SP, Brazil

Advisors: Dr. Flavio J. Caires & Dr. Massao Ionashiro

Master's in Chemistry
2013 - 2015

Institute of Chemistry, Federal University of Goiás — UFG · Goiânia, GO, Brazil

Advisor: Dr. Elias Y. Ionashiro

Licentiate Degree in Chemistry
2009 - 2013

Federal University of Mato Grosso — UFMT · Pontal do Araguaia, MT, Brazil

Advisor: Dr. George Barbosa da Silva